

## UNIT 2

### 2.1 NEED OR IDENTIFICATION OF A PROBLEM

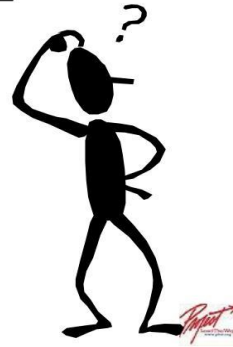
- Problem identification is a multi-step method that pain points and solves the problems faced by the users.
- With problem identification, you identify the root cause of a problem, develop a detailed problem statement and implement an ideal solution.



#### Problem Identification

##### Questions to Answer...

- Who ? - would need it
- What ? - is needed
- When ? - is it needed
- Where ? - is it needed
- Why ? - is it needed
- How Many ? - are needed



- The problem identification process allows you to look deeply into the issues at hand. And it encourages you to create a solution that directly targets the problem at its source.
- The world is full of entrepreneurs bursting with ideas. But for these ideas to be successful, they must address a clear need, solving for it at its root.

This stage is about understanding the needs of your target users and identifying problems they face. Conduct user research to gather insights about their habits, preferences. This can involve surveys, interviews, and observation.

## **1. Market Research**

Analyze the market to understand the competitive landscape, identify trends, and assess the potential for your solution. This involves examining existing products and services, analyzing industry data, and identifying market gaps

## **2. User Research**

Gather data about the target audience to understand their needs and expectations. Conduct surveys, interviews, and observation to gain insights into user behavior.

### **Define the problem**

Clearly articulate the issue or challenge.

**Example:** "There is a need for a portable CPR device that can assist untrained individuals in administering CPR during emergencies."

### **Analyze Current Solutions**

**Evaluate if there are existing solutions. If yes, identify the limitations.**

- Are they too expensive?
- Are they too complex for laypeople to use?
- Are they accessible in public spaces?

### **Assess the Impact**

**Quantify how addressing this problem could benefit the users or society.**

Example: "A portable, easy-to-use CPR device could save hundreds of lives each year, particularly in remote areas or situations where immediate medical help is unavailable."

## **Identify the user needs—what are people struggling with?**

For example:

- Many bystanders are reluctant to perform CPR due to lack of confidence.
- A device that guides and assists in CPR could help overcome this hesitation.

## **2.2 MARKET SURVEY:**

A market survey is conducted to gather real-world data from potential users and stakeholders. It helps validate the need, assess the demand, and refine the problem definition.

### **Purpose of a market survey**

- Understand customer needs and preferences.
- Evaluate market demand.
- Assess customer satisfaction.
- Identify market trends.
- Benchmark against competitors.
- Test marketing strategies.
- Determine pricing strategies.
- Support business decisions

A **market survey methodology** outlines the approach to collecting, analyzing, and interpreting data about a market, its customers, and competitors. Here's a step-by-step guide to an effective methodology

### **1. Define Objectives**

- Clearly outline the purpose of the survey.

*Examples:*

- Understand customer needs and preferences.

- Assess demand for a new product.
- Study market trends or competition.

## 2. Define Target Audience

- Identify who the survey is aimed at:
- Demographics: Age, gender, income, education, etc.
- Geographic location.
- Behavioural traits or purchasing habits.

## 3. Choose a Survey Type

- **Qualitative Surveys:** Open-ended questions to explore ideas in depth (e.g., focus groups, interviews).
- **Quantitative Surveys:** Structured questionnaires to gather numerical data (e.g., online surveys, phone surveys).

## 4. Determine Sample Size and Sampling Method

Decide how many people will participate and how they will be selected:

- **Random Sampling:** Ensures every individual in the target population has an equal chance.
- **Stratified Sampling:** Divides the population into segments (e.g., by age or location) and samples from each.
- **Convenience Sampling:** Uses easily accessible participants.

## 5. Collect Data

- Implement the survey using the chosen methods and tools (e.g., Google Forms, SurveyMonkey, interviews).
- Ensure ethical considerations (e.g., confidentiality, informed consent).

## 6. Analyse Data

- Organize and clean data to remove errors or inconsistencies.
- Use statistical tools or software like Excel, SPSS, or Python for analysis.
- Identify patterns, trends, and key insights.

## 7. Interpret Results

- Relate findings to the survey's objectives.
- Identify actionable insights and potential opportunities or challenges.

## 8. Present Findings

- Use visuals (charts, graphs) for clarity.

Write a detailed report summarizing:

- Objectives.
- Methodology.
- Key findings.
- Recommendations.

## 9. Take Action

- Use insights to make informed decisions (e.g., product development, marketing strategies).

## 2.5 PROBLEM DEFINITION

### Gather Information

Collect relevant data, facts, and insights about the problem from various sources, including stakeholders, industry experts, and existing research.

### Identify Patterns

Analyze the information to uncover trends, correlations, and **potential root causes** that may be contributing to the problem.

### Target Audience

Clearly define your target audience, understanding their demographics, psychographics, needs, and pain points

- **Demographics**

Age, gender, location, income, education, and other quantifiable characteristics.

- **Needs**

The fundamental requirements that your solution must address.

- **Psychographics**

Values, beliefs, interests, personality traits, and lifestyle choices.

- **Pain Points**

The specific problems or frustrations that your target audience faces.

## Prioritize Factors

Determine the **most significant factors** that are shaping the problem, focusing on those that have the greatest impact.

## Defining the Problem

The problem statement should clearly and concisely articulate the specific issue or challenge that needs to be addressed, based on the research.

## Identifying Impacts

The statement should also outline the potential consequences or impacts of the problem, both in the short-term and long-term, to emphasize the importance of finding a solution.

## User-centric

Ensure the problem is framed from the user's perspective, highlighting their needs and challenges.

## Measurable

Include quantifiable aspects to your problem statement to allow for effective measurement and progress tracking.

|                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>Example:</i></b>                                                                                                                                                                                                                         |
| <i>Students often struggle to focus on studying at home due to noise, visual distractions, and uncomfortable setups. Creating a dedicated, distraction-free study space could help them concentrate better and improve their productivity.</i> |
| <b><i>Problem Definition</i></b>                                                                                                                                                                                                               |
| <i>The problem can be defined as:</i>                                                                                                                                                                                                          |
| <ul style="list-style-type: none"><li><i>"Students lack a designated study space at home that minimizes distractions and promotes focus."</i></li></ul>                                                                                        |

## **2.6 KIPLING METHOD**

The Kipling Method is a tool used for problem-solving and decision-making. It involves asking a series of questions to gain a deeper understanding of the problem and potential solutions.

### **1.What?**

*Clearly define the problem and its context.*

### **2.Why?**

*Explore the underlying causes and motivations behind the problem.*

### **3.When?**

*Determine the timing and frequency of the problem.*

### **4.Where?**

*Identify the specific location or environment where the problem occurs.*

### **5.Who?**

*Understand who is affected by the problem.*

### **6.How?**

*Consider the ways in which the problem arises.*

### 2.6.1 Example: Designing a Self-Watering Plant Pot

**1. Who** will use this product?

It's designed for anyone who wants to grow plants, especially those who don't have time to water them regularly, like busy professionals, students, or elderly people.

**2. What** does the product do?

This pot automatically waters the plant as needed, ensuring it gets just the right amount of moisture over time.

**3. Where** will this product be used?

Mostly in homes, offices, or other indoor environments. It might also work on balconies or small outdoor spaces.

**4. When** is this product most useful?

It's particularly helpful for people who tend to forget watering schedules or travel frequently and want their plants to stay healthy in their absence.

**5. Why** is this product important?

Plants need consistent moisture to thrive, and over- or under-watering can harm them. This product helps maintain the right level of hydration for healthy plant growth.

**6. How** will the product work?

The pot includes a water reservoir at the base. Through a simple capillary system or moisture sensor, it releases water only when the soil is dry, keeping the plant hydrated without manual watering.



## **2.6.2 Solve the following problem using the Kipling method:**

*“We create a sustainable, affordable, and accessible urban transportation system”*

### **1.What?**

#### **What is the problem or goal?**

To develop an urban transportation system that is sustainable, affordable, and accessible to all segments of the population.

### **2. Why?**

#### **Why is this important?**

- To reduce environmental impact (sustainability).
- To make transportation cost-effective for users (affordability).
- To ensure equitable access for people of all socioeconomic backgrounds, including individuals with disabilities (accessibility).
- To address urban issues like congestion, pollution, and inefficiency.

### **3. How**

#### **How can this be achieved?**

- Implement eco-friendly vehicles (electric buses, trams, or bikes).
- Use renewable energy sources for power (solar-powered charging stations).
- Develop smart ticketing systems with subsidies for low-income groups.
- Ensure infrastructure like ramps, elevators, and wider entrances for accessibility.
- Promote public-private partnerships to reduce costs.

- Use data analytics to optimize routes and schedules for efficiency.

#### **4. When?**

When should this be implemented?

- Short-term (0–1 year): Conduct feasibility studies, pilot projects, and community engagement.
- Medium-term (1–5 years): Build infrastructure, implement policies, and scale up pilot projects.
- Long-term (5+ years): Continuously upgrade systems and technologies, expand reach, and refine based on user feedback.

#### **5. Where?**

**Where will this system be implemented?**

- High-density urban areas where public transport demand is highest.
- Low-income neighbourhoods to ensure inclusivity.
- Key transit points like airports, railway stations, and commercial hubs to enhance connectivity.

#### **6. Who**

**Who will be involved?**

- Government authorities: For policymaking, funding, and regulation.
- Private sector partners: For innovation and implementation of solutions.
- Urban planners and engineers: To design efficient systems.
- Non-profit organizations: To advocate for accessibility and inclusivity.
- Residents: To participate in decision-making and provide feedback.

## **Actionable Plan**

By systematically addressing each aspect using the Kipling method, the transportation system can be developed to meet sustainability, affordability, and accessibility goals effectively.

### **2.7 CHALLENGE STATEMENT:**

*Formulate a question that makes it possible to ideate potential solutions in a targeted way.*

- Challenge statements are concise and focused descriptions that articulate specific problems, opportunities, or goals within a given context. These statements serve as the foundation for problem-solving, innovation, and strategic planning efforts.
- It is designed to focus efforts and inspire innovative solutions.
- By clearly defining the challenges at hand, challenge statements provide a shared understanding among individuals or teams, guiding them towards generating targeted solutions and ideas.
- These statements often address a gap, a need for improvement, or a desired outcome, prompting creative thinking and encouraging a systematic approach to finding effective solutions.
- Challenge statements play a crucial role in various fields, including design thinking, entrepreneurship, and research, as they help frame the problem space and drive focused efforts towards meaningful and actionable results.

To inspire innovation by framing the problem as an opportunity to create solutions. Encourages exploration of possibilities rather than dwelling on the negative aspects. Open-ended and optimistic, focusing on "how might we" or "how can we" questions. Often used in design thinking, brainstorming, and innovation processes.

## **2.8 Problem statement Initial Specification**

### **Scope and Boundaries**

Clearly define the problem's scope and what's included and excluded. Establishing clear boundaries is crucial for focused problem-solving.

### **Constraints and Requirements**

Identify any limitations, such as budget, time, or resources. Understand the specific requirements that need to be met for a successful solution.

### **Target Audience and Users**

Consider the intended users and their needs. Define the target audience and their expectations for the problem's solution

### **Desired Outcomes and Metrics**

Establish clear goals for the solution. Identify measurable metrics to evaluate the success of the creative process and its outcomes.

**2.9 Problem-solving techniques** are systematic approaches used to identify, analyse, and resolve challenges effectively. These methods are crucial in various fields, including engineering, education, business, and daily life, to ensure logical and creative solutions.

**Problem-solving is the process of:**

1. Identifying and defining a problem.
2. Analysing its root causes.
3. Generating potential solutions.
4. Evaluating and implementing the best solution.
5. Monitoring outcomes to ensure the problem is resolved.

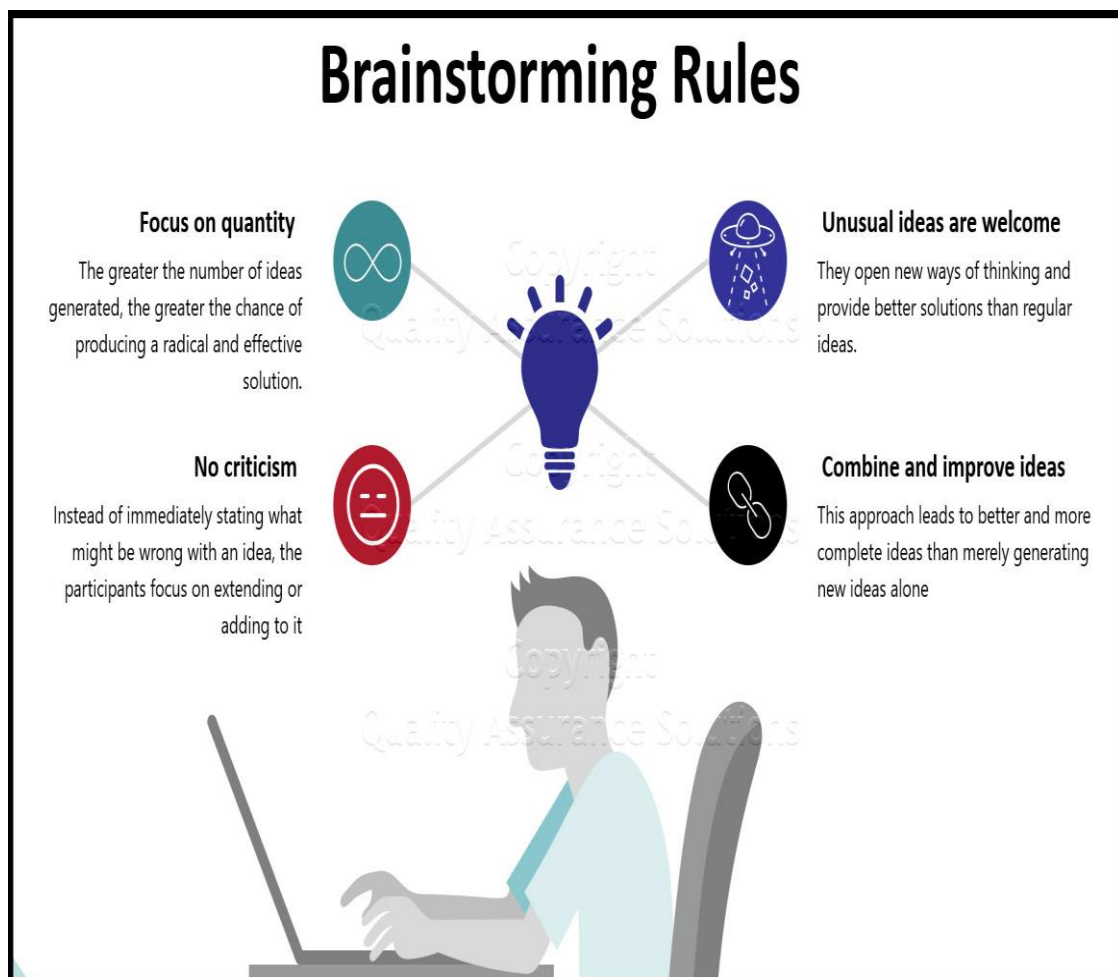
There are various techniques depending on the problem's nature:

1. Brain storming
2. Analogy technique or synectics
3. Check list
4. Trigger words
5. Morphological method
6. Interaction matrix method

## 2.10 BRAINSTORMING

Brainstorming is a creative thinking technique for coming up with new ideas and solving problems. Teams use this ideation method to encourage new ways of thinking and collectively generate solutions.

Brainstorming encourages free thinking and allows for all ideas to be voiced without judgment, fostering an open and innovative environment. This process typically involves a group of people, although it can be done individually as well.



### **2.10.1 Purpose Of Brainstorming**

The primary purpose of a brainstorming session is to generate and document many ideas, no matter how “out there” they might seem. Through this lateral thinking process, inventive ideas are suggested, which sparks creative solutions. By encouraging everyone to think more freely and not be afraid to share their ideas, teams can build on each other’s thoughts to find the best possible solution to a problem.

Brainstorming usually takes place in a group setting where people get together to creatively solve problems and come up with ideas. However, it’s also useful for individuals who need to explore novel solutions to a problem. Sitting down by yourself and writing down solutions to potential problems is a great way to brainstorm individually. Focusing your mind on a defined problem allows you to think of many creative ways to get to an answer.

While brainstorming normally allows for free-form methods of thinking and doesn’t require many rules, the best results usually stem from controlled sessions. Posing questions and role-playing different scenarios during the brainstorming session is a smart way to pull out unusual ideas and never-before-thought-of solutions.

### **2.10.2 Benefits of brainstorming**

- Encourages creativity
- Fosters collaboration and team building
- Generates innovative, revolutionary ideas
- Establishes different perspectives
- Introduces many ideas quickly

## 2.11 SYNECTIC'S: Unleashing Creativity Through Analogy

- Synectic's is a powerful problem-solving technique that taps into the human mind's natural ability to *draw connections between seemingly unrelated concepts*.
- By harnessing the power of analogies, Synectic's helps individuals and teams generate innovative solutions to complex challenges.

### 2.11.1 The Synectic's Process:

#### 1. Understand the Problem

Clearly define the problem or challenge at hand, ensuring a shared understanding among the team.

#### 2. Generate Analogies

Explore analogies, both direct and personal, to uncover new perspectives and spark creative thinking.

#### 3. Develop Solutions

Use the insights gained from the analogies to generate innovative solutions to the original problem.

### 2.11.2 Generating Ideas through Analogies

| Direct Analogy                                                                         | Personal Analogy                                                                                  | Symbolic Analogy                                                                                      |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Drawing <i>comparisons</i> between the problem and other existing solutions or systems | Imagining <i>oneself as the problem or solution</i> , fostering empathy and deeper understanding. | Representing the problem or solution through <i>metaphors and symbols</i> to uncover hidden meanings. |



### 2.11.3 Applying Synectics to a Problem:

#### Define the Problem

Clearly articulate the challenge or opportunity you wish to address.

#### Generate Analogies

Explore direct, personal, and symbolic connections to uncover new perspectives.

#### Develop Solutions

Translate the insights gained from the analogies into innovative solutions

#### Benefits of Synectics:

- Synectics *encourages the use of analogies* to uncover new perspectives and generate innovative solutions.
- Fostering an environment that *suspends judgment and encourages unusual connections* is key to the Synectics process.
- The Synectics method thrives in a *collaborative setting, where diverse perspectives and ideas* can be shared and built upon.

### 2.11.4 Example Problem: Designing a Better Umbrella

|                                                                                                                                  |                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Compact Design</u></b><br>Develop an umbrella that can fold down to a smaller size for easy portability.                   | <b><u>Self-Deploying</u></b><br>Incorporate a mechanism that allows the umbrella to open and close with the push of a button.                   |
| <b><u>Hands-Free</u></b><br>Explore a design that frees up the user's hands, such as a shoulder-mounted or head-mounted umbrella | <b><u>Weather-Resistant</u></b><br>Enhance the umbrella's durability and weather-resistant properties to withstand strong winds and heavy rain. |

## **Synectics Solution Walkthrough**

### **Direct Analogy**

Observe how a telescope collapses into a compact size, then apply this concept to the umbrella design.

### **Personal Analogy**

Imagine wearing the umbrella on your head, leading to a hands-free, wearable design.

### **Symbolic Analogy**

Represent the umbrella as a shield against the elements, inspiring weather-resistant features.

### **Automated Deployment**

Draw inspiration from the push-button mechanisms used in other devices to create a self-deploying umbrella.

## **2.12 THE CHECKLIST TECHNIQUE**

Checklist is a list of alternative questions you can ask yourself about the problem. It serves to remind you of other ways of looking at the problem so that new imagination stimuli can be introduced to the idea generation process. Osborn's checklist is widely used as a model.

- A series of simple questions, which can be used either individually or in groups, designed to support creative and divergent thinking when faced by a design problem.
- The questions need a point of focus, which could either be an existing solution or proposed concepts to a design problem. The questions should be taken one at a time, to explore new ways and approaches to the problem

## CHECKLIST

|             |                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Other uses? | New ways to use as is? Other uses if modified?                                                                                                                   |
| Adapt?      | What else is like this? What other idea does this suggest? Does past offer parallel? What could I copy? Whom could I emulate?                                    |
| Modify?     | New twist? Change meaning, colour, motion, odour, taste, form, shape? Other changes?                                                                             |
| Magnify?    | What to add? More time? Greater frequency? Stronger? Higher? Larger? Longer? Thicker? Heavier? Extra value? Plus ingredient? Duplicate? Multiply? Exaggerate?    |
| Minify?     | What to subtract? Smaller? Condensed? Miniature? Lower? Shorter? Narrower? Lighter? Omit? Streamline? Split up? Understate? Less frequent?                       |
| Substitute? | Who else instead? What else instead? Other ingredient? Other material? Other process? Other power? Other place? Other approach? Other tone of voice? Other time? |
| Rearrange?  | Interchange components? Other pattern? Other layout? Other sequence? Transpose cause and effect? Change place? Change schedule? Earlier? Later?                  |
| Reverse?    | Transpose positive and negative? How about opposites? Turn it backward, upside down, inside out? Reverse roles? Change shoes? Turn tables? Turn other cheek?     |
| Combine?    | How about a blend, an alloy, an assortment, an ensemble? Combine units?                                                                                          |

### 2.12.1 EXAMPLE:

*"How can we improve the efficiency of a water filtration system?"*

#### 1. Put to other uses

**Question:** Can the filtration system serve a purpose beyond just water filtration?

**Example:** Design the system to also filter air or extract minerals from water for industrial purposes.

#### 2. Adapt

**Question:** What techniques or ideas from other industries can be applied?

**Example:** Use reverse osmosis technology borrowed from desalination plants to improve filtration precision in household systems.

#### 3. Modify

**Question:** How can the system's design, structure, or process be adjusted?

**Example:** Introduce a compact, modular design that allows users to add or remove filtration stages based on water quality.

#### 4. Magnify

**Question:** How can the filtration system be enhanced or expanded?

**Example:** Add sensors to monitor water quality in real time and alert users when maintenance or filter replacement is needed.

#### 5. Minify

**Question:** How can the system be simplified or reduced?

**Example:** Develop a portable, lightweight version for use in emergencies or outdoor activities.

#### 6. Substitute

**Question:** What materials or processes can be replaced?

**Example:** Replace traditional carbon filters with graphene-based filters for faster and more efficient filtration.

## 7. Rearrange

**Question:** Can the order or configuration of the system be changed?

**Example:** Rearrange the filtration stages to prioritize removing large contaminants before finer filtration, improving overall efficiency and extending filter life.

## 8. Reverse

**Question:** Can the process flow be reversed or reimagined?

**Example:** Create a system that captures wastewater from filtration for secondary use, like irrigation, instead of discarding it.

## 9. Combine

**Question:** Can this system be integrated with other technologies?

**Example:** Combine the filtration system with a renewable energy source like solar panels to make it energy-efficient and suitable for off-grid areas.

## Final Result:

By applying Osborn's Checklist, possible innovations include:

- A **graphene-based filtration system** with modular design for customizable water treatment.
- Real-time **water quality sensors** integrated with an app for user notifications.
- A **solar-powered portable filtration unit** for remote locations.
- A system that **repurposes wastewater** for non-potable uses, improving sustainability.

This process demonstrates how the checklist can systematically explore new ideas and optimize technical solutions.

## 2.13 TRIGGER WORDS:

The **Trigger Words Problem-Solving Technique** is a creativity tool that involves using specific words or phrases to generate ideas and solutions for a problem. These words "trigger" new ways of thinking by prompting alternative perspectives and actions.

### Identify the Problem:

- Clearly define the issue you want to solve.

### Select Relevant Trigger Words:

- Use a list of action-oriented words as prompts.
- Common trigger words include: **Expand, Reduce, Replace, Combine, Adapt, Optimize, Shift, Simplify, Transform, Modify.**

#### 2.13.1 Problem: *Improving the durability and performance of electric vehicle (EV) batteries.*

### Trigger Words in Action

#### 1.Expand

Expand the battery's ability to handle extreme temperatures for better performance in diverse climates.

Expand its application by enabling batteries to power home energy systems when vehicles are idle.

#### 2.Reduce

Reduce charging time by incorporating ultra-fast charging technology.

Reduce weight by using lightweight materials like graphene or solid-state technology.

#### 3.Replace

Replace liquid electrolytes with solid-state alternatives to reduce the risk of overheating and fires.

Replace cobalt with more sustainable and less expensive materials like lithium iron phosphate (LFP).

#### **4.Combine**

Combine the battery system with a regenerative braking mechanism to recharge while driving.

Combine solar panel integration with the car's roof to provide supplementary charging.

#### **5.Simplify**

Simplify the design for easier maintenance and repair, such as modular battery packs.

Simplify charging by implementing universal connectors across EV brands.

#### **6.Speed**

Speed up innovation cycles by using AI for battery testing and optimization.

Speed up charging with wireless charging pads integrated into parking spots.

#### **7.Optimize**

Optimize energy density to allow smaller batteries with the same capacity.

Optimize battery management systems with AI to balance performance and longevity.

#### **8.Adapt**

Adapt battery designs for second-life applications, such as stationary storage in homes.

Adapt the system for extreme environments, such as military or space applications.

#### **9.Transform**

Transform the battery into a **bidirectional energy source**, allowing it to send energy back to the grid during peak demand (vehicle-to-grid technology).

Transform the battery into a circular system where materials are 100% recyclable.

## 2.14 MORPHOLOGICAL METHOD

The morphological method, or morphological analysis, is a structured problem-solving technique especially useful for complex and multi-dimensional problems. This method systematically breaks down a problem into its fundamental attributes, then explores potential solutions by analyzing all possible combinations of these attributes.

**Define the Problem:** Clearly state the problem or challenge you are addressing. This method is most effective for issues that are not straightforward and have many influencing factors.

**Identify Key Attributes:** Break the problem down into its essential components or parameters. For example, if designing a new portable CPR device, the key attributes might be power source, portability, interface, and materials.

**List Options for Each Attribute:** For each parameter, list as many possible solutions or variations as possible. For example, under "power source," you might list battery, solar, and manual operation.

**Create a Morphological Matrix:** Arrange the parameters along one axis and the possible solutions along another, creating a grid. Each cell in the grid represents a unique combination of solutions.

**Analyze Combinations:** Systematically explore different combinations in the matrix to find novel solutions. Some combinations may not be feasible, but others might lead to innovative ideas or previously unconsidered solutions.

**Evaluate and Refine:** Assess each promising combination based on feasibility, cost, and other criteria relevant to the problem. Select the best solution or refine the top combinations for further development.



### 2.14.1 Problem: Designing an Eco-Friendly Urban Vehicle

We want to create a small, sustainable vehicle tailored for city environments. Key parameters for this vehicle could include fuel type, vehicle size, materials, and user interface.

| Parameter             | Option 1                | Option 2              | Option 3                    |
|-----------------------|-------------------------|-----------------------|-----------------------------|
| <i>Fuel Type</i>      | <i>Electric</i>         | <i>Hydrogen</i>       | <i>Solar</i>                |
| <i>Vehicle Size</i>   | <i>One-seater</i>       | <i>Two-seater</i>     | <i>Cargo space included</i> |
| <i>Materials</i>      | <i>Recycled plastic</i> | <i>Aluminium</i>      | <i>Bamboo or wood</i>       |
| <i>User Interface</i> | <i>Touchscreen</i>      | <i>Simple buttons</i> | <i>App-based control</i>    |

**Analyze Combinations:** A few possible combinations to consider might include:

**Combination 1:** *Electric, one-seater, recycled plastic, app-based control*

Ideal for single commuters looking for an eco-friendly personal transport option.

**Combination 2:** *Solar, two-seater, bamboo, touchscreen*

An ultra-sustainable option targeting couples or friends.

**Combination 3:** *Hydrogen, cargo space included, aluminum, simple buttons*

Suitable for light urban deliveries, with a focus on sustainability and efficiency.

#### **Evaluate and Refine:**

After identifying interesting combinations, the next step would involve evaluating these designs based on cost, practicality, and potential environmental impact. This might involve prototyping or conducting surveys with potential users to refine the chosen designs.

## 2.15 INTERACTION MATRIX METHOD

The interaction matrix method is a useful tool for problem-solving, particularly in understanding *how different elements or variables within a system interact with each other*.

**Interaction Matrix:**

*Example: Designing an Electric Vehicle (EV)*

| Component       | Battery | Electric Motor | Braking System | Charging System |
|-----------------|---------|----------------|----------------|-----------------|
| Battery         | No      | Strong         | Weak           | Strong          |
| Electric Motor  | Strong  | No             | Moderate       | Weak            |
| Braking System  | Weak    | Moderate       | No             | Weak            |
| Charging System | Strong  | Weak           | Weak           | No              |

**Explanation:**

- **Battery & Electric Motor:** Strong Interaction. The battery powers the electric motor. The battery's charge and capacity directly affect the motor's performance (e.g., how far the vehicle can travel).
- **Battery & Charging System:** Strong Interaction. The charging system is responsible for replenishing the battery's energy. The charging speed and method (e.g., fast charging) are important for battery health.
- **Electric Motor & Braking System:** Moderate Interaction. When braking, especially with regenerative braking, the motor helps recover energy and sends it back to the battery. So, while the motor influences braking, the interaction is moderate compared to battery-motor relations.

- **Braking System & Charging System:** Weak Interaction. While braking can help regenerate energy (and feed it back into the battery), the braking system doesn't directly affect the charging system.

This simplified interaction matrix helps prioritize the battery, electric motor, and charging system as the key components to focus on. The battery and motor have a strong interaction, so ensuring they work well together is crucial for EV performance. Similarly, the charging system and battery are strongly linked, so efficient charging must be prioritized.